



Solids

A SOLID IS A THREE-DIMENSIONAL SHAPE.

Solids are objects with three dimensions: width, length, and height. They also have surface areas and volumes.

Prisms

Many common solids are polyhedrons—three-dimensional shapes with flat surfaces and straight edges. Prisms are a type of polyhedron made up of two parallel shapes of exactly the same shape and size, which are connected by faces. In the example to the right, the parallel shapes are pentagons, joined by rectangular faces. Usually a prism is named after the shape of its ends (or bases), so a prism whose parallel shapes are rectangles is known as a rectangular prism. If all its edges are equal sizes, it is called a cube.

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◀ 134–137 Polygons

Volumes 154–155 ▶

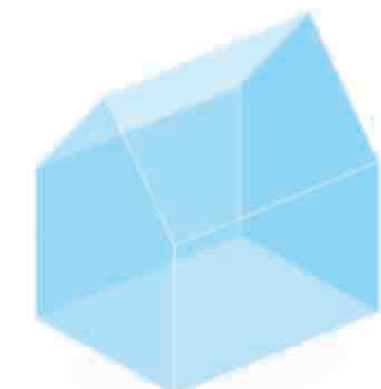
Surface area of solids 156–157 ▶

▷ A prism

The cross section of this prism is a pentagon (a shape with five sides), so it is called a pentagonal prism.

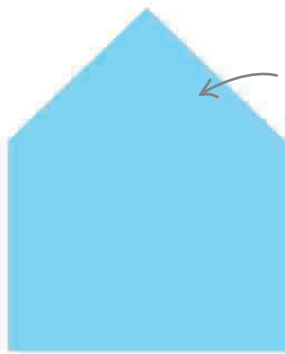
◁ Volume

The amount of space that a solid occupies is called its volume.



cross section of pentagonal prism is a pentagon

a pentagon is a shape with five sides

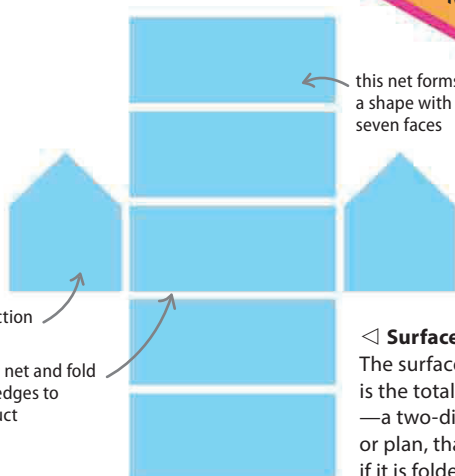


△ Cross section

A cross section is the shape made when an object is sliced from top to bottom.

cross section

cut out net and fold along edges to construct



this net forms a shape with seven faces

◁ Surface area

The surface area of a solid is the total area of its net—a two-dimensional shape, or plan, that forms the solid if it is folded up.

